Main Category	Category	Terms
Disease	OMIM_DISEASE	Hypoparathyroidism, sensorineural deafness, and renal dysplasia
Disease	UP_KW_DISEASE	Deafness
Disease	UP_KW_DISEASE	Disease variant
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Immunity
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Innate immunity
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Activator
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	MOTIF:YxKxHxxxRP
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Flexible linker
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with TBX21
Functional Annotations	UP_SEQ_FEATURE	ZN_FING:GATA-type 1
Functional Annotations	UP_SEQ_FEATURE	ZN_FING:GATA-type 2
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	cell morphogenesis
Gene	GOTERM_BP_DIRECT	in utero embryonic development

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	cell fate determination
Gene Ontology	GOTERM_BP_DIRECT	neuron migration
Gene Ontology	GOTERM_BP_DIRECT	cell activation
Gene Ontology	GOTERM_BP_DIRECT	type IV hypersensitivity
Gene Ontology	GOTERM_BP_DIRECT	regulation of cytokine production
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cytokine production
Gene Ontology	GOTERM_BP_DIRECT	kidney development
Gene Ontology	GOTERM_BP_DIRECT	mesonephros development
Gene Ontology	GOTERM_BP_DIRECT	lens development in camera-type eye
Gene Ontology	GOTERM_BP_DIRECT	immune system process
Gene Ontology	GOTERM_BP_DIRECT	immune system development
Gene Ontology	GOTERM_BP_DIRECT	pro-T cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	aortic valve morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	cardiac right ventricle morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	ventricular septum development
Gene Ontology	GOTERM_BP_DIRECT	chromatin remodeling
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	defense response
Gene	GOTERM_BP_DIRECT	inflammatory response

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	humoral immune response
Gene Ontology	GOTERM_BP_DIRECT	signal transduction
Gene Ontology	GOTERM_BP_DIRECT	nervous system development
Gene Ontology	GOTERM_BP_DIRECT	axon guidance
Gene Ontology	GOTERM_BP_DIRECT	cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	male gonad development
Gene Ontology	GOTERM_BP_DIRECT	response to xenobiotic stimulus
Gene Ontology	GOTERM_BP_DIRECT	response to virus
Gene Ontology	GOTERM_BP_DIRECT	anatomical structure morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	post-embryonic development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of signal transduction
Gene Ontology	GOTERM_BP_DIRECT	response to gamma radiation
Gene Ontology	GOTERM_BP_DIRECT	gene expression
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of endothelial cell migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of epithelial to mesenchymal transition
Gene Ontology	GOTERM_BP_DIRECT	regulation of neuron projection development
Gene Ontology	GOTERM_BP_DIRECT	cell differentiation
Gene	GOTERM_BP_DIRECT	neuron differentiation

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	T cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	erythrocyte differentiation
Gene Ontology	GOTERM_BP_DIRECT	macrophage differentiation
Gene Ontology	GOTERM_BP_DIRECT	regulation of epithelial cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	TOR signaling
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of type II interferon production
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of interleukin-2 production
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of interleukin-13 production
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of interleukin-4 production
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of interleukin-5 production
Gene Ontology	GOTERM_BP_DIRECT	T cell differentiation in thymus
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of mammary gland epithelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	embryonic hemopoiesis
Gene Ontology	GOTERM_BP_DIRECT	cellular response to interferon-alpha
Gene Ontology	GOTERM_BP_DIRECT	ureter maturation
Gene Ontology	GOTERM_BP_DIRECT	parathyroid hormone secretion
Gene Ontology	GOTERM_BP_DIRECT	norepinephrine biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	inner ear morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol 3-kinase/protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	regulation of neuron apoptotic process
Gene	GOTERM_BP_DIRECT	ear development

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	response to estrogen
Gene Ontology	GOTERM_BP_DIRECT	thymic T cell selection
Gene Ontology	GOTERM_BP_DIRECT	T-helper 2 cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	innate immune response
Gene Ontology	GOTERM_BP_DIRECT	cell fate commitment
Gene Ontology	GOTERM_BP_DIRECT	response to ethanol
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of T cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	regulation of T-helper cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell cycle
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	cell maturation
Gene Ontology	GOTERM_BP_DIRECT	sympathetic nervous system development
Gene Ontology	GOTERM_BP_DIRECT	thymus development
Gene Ontology	GOTERM_BP_DIRECT	embryonic organ development
Gene Ontology	GOTERM_BP_DIRECT	developmental growth
Gene Ontology	GOTERM_BP_DIRECT	anatomical structure formation involved in morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	system development
Gene	GOTERM_BP_DIRECT	homeostasis of number of cells

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of inflammatory response
Gene Ontology	GOTERM_BP_DIRECT	regulation of developmental process
Gene Ontology	GOTERM_BP_DIRECT	T cell receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	cartilage development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of phosphatidylinositol 3-kinase/ protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	parathyroid gland development
Gene Ontology	GOTERM_BP_DIRECT	pharyngeal system development
Gene Ontology	GOTERM_BP_DIRECT	uterus development
Gene Ontology	GOTERM_BP_DIRECT	canonical Wnt signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	mesenchymal to epithelial transition
Gene Ontology	GOTERM_BP_DIRECT	mast cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	ureteric bud formation
Gene Ontology	GOTERM_BP_DIRECT	cellular response to cytokine stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to interleukin-4
Gene Ontology	GOTERM_BP_DIRECT	cellular response to tumor necrosis factor
Gene Ontology	GOTERM_BP_DIRECT	otic vesicle development
Gene Ontology	GOTERM_BP_DIRECT	cellular response to BMP stimulus
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of ureteric bud formation
Gene Ontology	GOTERM_BP_DIRECT	nephric duct morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	nephric duct formation
Gene	GOTERM_BP_DIRECT	regulation of nephron tubule epithelial cell

Main Category	Category	Terms
Ontology		differentiation
Gene Ontology	GOTERM_BP_DIRECT	ureter morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	lymphocyte migration
Gene Ontology	GOTERM_BP_DIRECT	cochlea development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of miRNA transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of establishment of cell polarity
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell motility
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of endothelial cell apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of T-helper 2 cell cytokine production
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell proliferation involved in mesonephros development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of thyroid hormone generation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription regulatory region DNA binding
Gene Ontology	GOTERM_BP_DIRECT	regulation of cellular response to X-ray
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of fibroblast growth factor receptor signaling pathway involved in ureteric bud formation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of glial cell-derived neurotrophic factor receptor signaling pathway involved in ureteric bud formation
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_MF_DIRECT	transcription cis-regulatory region binding
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II transcription regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II cis-regulatory region sequence-specific DNA binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity, RNA polymerase II-specific
Gene Ontology	GOTERM_MF_DIRECT	cis-regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	transcription coactivator binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription repressor activity, RNA polymerase II-specific
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription activator activity, RNA polymerase II-specific
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity
Gene Ontology	GOTERM_MF_DIRECT	interleukin-2 receptor binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	zinc ion binding
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	metal ion binding
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II-specific DNA-binding transcription factor binding
Gene Ontology	GOTERM_MF_DIRECT	E-box binding
Gene Ontology	GOTERM_MF_DIRECT	HMG box domain binding
Gene Ontology	GOTERM_MF_DIRECT	histone methyltransferase binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific double-stranded DNA binding
General Annotations	CHROMOSOME	10
General Annotations	CYTOGENETIC_LOCATION	10p14
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes a protein which belongs to the GATA family of transcription factors. The protein contains two GATA-type zinc fingers and is an important regulator of

Main Category	Category	Terms
		T-cell development and plays an important role in endothelial cell biology. Defects in this gene are the cause of hypoparathyroidism with sensorineural deafness and renal dysplasia. [provided by RefSeq, Nov 2009]
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=2; Name=1; IsoId=P23771-1; Sequence=Displayed; Name=2; IsoId=P23771-2; Sequence=VSP_001598;
General Annotations	UP_COMMENT_DISEASE	Hypoparathyroidism, sensorineural deafness, and renal disease (HDR) [MIM:146255]: A disease characterized by steroid-resistant nephrosis with progressive renal failure, hypoparathyroidism, sensorineural deafness, and renal dysplasia. {ECO:0000269|PubMed:10935639, ECO:0000269|PubMed:11389161, ECO:0000269|PubMed:26514990}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	Binds DNA via the 2 GATA-type zinc fingers. Each zinc finger may bind either adjacent sites in a palindromic motif, or a different DNA molecule allowing looping and long-range gene regulation.
General Annotations	UP_COMMENT_DOMAIN	The YxKxHxxxRP motif is critical for DNA-binding and function. {ECO:0000250}.
General Annotations	UP_COMMENT_FUNCTION	Transcriptional activator which binds to the enhancer of the T-cell receptor alpha and delta genes. Binds to the consensus sequence 5'-AGATAG-3'. Required for the T-helper 2 (Th2) differentiation process following immune and inflammatory responses. Positively regulates ASB2 expression (By similarity). Coordinates macrophage transcriptional activation and UCP2-dependent metabolic reprogramming in response to IL33. Upon tissue injury, acts downstream of IL33 signaling to drive differentiation of inflammation-resolving alternatively activated macrophages. {ECO:0000250|UniProtKB:P23772, ECO:0000269|PubMed:23824597}.
General Annotations	UP_COMMENT_INTERACTION	P23771; P35609: ACTN2; NbExp=3; IntAct=EBI-6664760, EBI-77797; P23771; P28329-3: CHAT; NbExp=3; IntAct=EBI-6664760, EBI-25837549; P23771; Q96KQ7: EHMT2; NbExp=20; IntAct=EBI-6664760, EBI-744366; P23771; P22607: FGFR3; NbExp=3; IntAct=EBI-6664760, EBI-348399; P23771; Q9NP62: GCM1; NbExp=9; IntAct=EBI-6664760, EBI-21194843; P23771; O15379: HDAC3; NbExp=7; IntAct=EBI-6664760, EBI-607682; P23771; Q9UQL6: HDAC5; NbExp=4; IntAct=EBI-6664760, EBI-715576; P23771; Q16665: HIF1A; NbExp=3; IntAct=EBI-6664760, EBI-447269; P23771; P06239: LCK; NbExp=3; IntAct=EBI-6664760, EBI-1348; P23771; Q9BTC8: MTA3; NbExp=18; IntAct=EBI-6664760, EBI-2461787; P23771; P25788: PSMA3; NbExp=3; IntAct=EBI-6664760, EBI-348380; P23771; Q9UL17: TBX21; NbExp=6; IntAct=EBI-6664760, EBI-3922312; P23771-2; A8MQ03: CYSRT1; NbExp=3; IntAct=EBI-6672518, EBI-3867333; P23771-2; P59990: KRTAP12-1; NbExp=3; IntAct=EBI-6672518, EBI-10210845; P23771-2; Q52LG2:

Main Category	Category	Terms
		KRTAP13-2; NbExp=3; IntAct=EBI-6672518, EBI-11953846; P23771-2; Q3LI66: KRTAP6-2; NbExp=3; IntAct=EBI-6672518, EBI-11962084; P23771-2; Q3LI67: KRTAP6-3; NbExp=3; IntAct=EBI-6672518, EBI-22311199; P23771-2; Q58EX7: PLEKHG4; NbExp=3; IntAct=EBI-6672518, EBI-949255;
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus.
General Annotations	UP_COMMENT_SUBUNIT	Interacts with TBX21 ('Tyr-530' phosphorylated form). {ECO:0000269|PubMed:15662016}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	T-cells and endothelial cells.
Identifiers	ENTREZ_GENE_ID	2625
Identifiers	ENSEMBL_GENE_ID	ENSG00000107485
Identifiers	UNIPROT_ID	GATA3_HUMAN
Identifiers	UNIPROT_ID	A0A994J6H6_HUMAN
Identifiers	UNIPROT_ID	A0A2R8Y2A9_HUMAN
Identifiers	UNIPROT_ID	A0A2R8Y4T2_HUMAN
Interactions	UP_KW_LIGAND	Metal-binding
Interactions	UP_KW_LIGAND	Zinc
Literature	GENERIF_SUMMARY	Strong GATA-3 binding to an oligonucleotide containing two GATA-3 motifs has been mapped at a DNase hypersensitivity site in cluster IV of the CD8 gene complex.
Literature	GENERIF_SUMMARY	Identification of an alternative GATA-3 promoter directing tissue-specific gene expression
Literature	GENERIF_SUMMARY	A marked reduction in both GATA-3 mRNA and protein expression in HPV-immortalized cell lines was confirmed in cervical carcinoma cell lines.
Literature	GENERIF_SUMMARY	GATA-3 transcriptional imprinting in Th2 lymphocytes
Literature	GENERIF_SUMMARY	VZV virus and mite antigens induce expression of this transcription factor in cord blood mononuclear cells, differenetially affecting Th1 and Th2 cells
Literature	GENERIF_SUMMARY	GATA-3 is an important transcription factor in regulating human Th2 cell differentiation in vivo.
Literature	GENERIF_SUMMARY	GATA3 mutations may have a role in the hypoparathyroidism, deafness, and renal dysplasia (HDR) syndrome
Literature	GENERIF_SUMMARY	GATA3 plays important roles in the maintenance of the Th2 phenotype and continuous chromatin remodeling of the specific Th2 cytokine gene locus through cell division.
Literature	GENERIF_SUMMARY	GATA-3 suppresses IFN-gamma promoter activity

Main Category	Category	Terms
		independently of binding to cis-regulatory elements
Literature	GENERIF_SUMMARY	Erythropoietin (Epo) gene expression is under the control of hypoxia-inducible factor 1 (HIF-1), and is negatively regulated by GATA3.
Literature	GENERIF_SUMMARY	GATA3 is involved in growth control and the maintenance of the differentiated state in epithelial cells and may contribute to tumorigenesis in breast cancer.
Literature	GENERIF_SUMMARY	GATA-3 contributed to the production of IL-4, IL-5 in patients with allergic rhinitis.
Literature	GENERIF_SUMMARY	Expressed in Hodgkin disesase Reed Sternberg cells.
Literature	GENERIF_SUMMARY	GATA-3 binds and transactivates the NKG2A promoter; coexpression of human GATA-3 with an NKG2A promoter construct in K562 cells leads to enhanced promoter activity.
Literature	GENERIF_SUMMARY	interleukin-5 transcription repression by the glucocorticoid receptor targets GATA3 signaling and involves histone deacetylase recruitment
Literature	GENERIF_SUMMARY	GATA3 is crucially involved in IL-5 gene transcription in human peripheral CD4-positive t cells.
Literature	GENERIF_SUMMARY	strong specific upregulation of Gata-3 impaired nuclear translocation and its cooperative action with the TGF-beta pathway, suggesting that Gata-3 plays a central role in human pancreatic cancer
Literature	GENERIF_SUMMARY	DL1-induced activation of the Notch1 pathway controls the lineage commitment of early thymic precursors by altering the levels between Spi-B and GATA-3.
Literature	GENERIF_SUMMARY	Low expression of GATA3 is associated with invasive breast carcinomas
Literature	GENERIF_SUMMARY	Segmental allergen challenge in asthmatics leads to increased GATA-3, c-maf and T-bet expression in BAL cells but not in bronchial biopsies
Literature	GENERIF_SUMMARY	Direct sequencing after PCR amplification of genomic DNA revealed a novel insertional mutation (405insC) in the GATA3 gene of two patients with HDR syndrome (Hypoparathyroidism, sensorineural Deafness and Renal anomaly syndrome).
Literature	GENERIF_SUMMARY	Three novel mutations in the GATA3 gene responsible for familial hypoparathyroidism and deafness.
Literature	GENERIF_SUMMARY	Imbalance of transcription factors T-bet and GATA-3 may be one of the key factors in immune dysregulation of recurrent aphthous ulcerations.
Literature	GENERIF_SUMMARY	the GATA-3/T-bet transcription factor complex regulates the cell-lineage-specific expression of the lymphocyte homing receptors
Literature	GENERIF_SUMMARY	GATA3 is probably a mediator for the transcriptional

Main Category	Category	Terms
		upregulation of MUC1 expression in some breast cancers.
Literature	GENERIF_SUMMARY	Overexpression of GATA3 rescues T-helper type 2 (Th2) cytokine expression in protein kinase C theta-deficient transgenic Th2 cells.
Literature	GENERIF_SUMMARY	A novel heterozygous deletion frameshift mutation of GATA3 in a Japanese kindred with the hypoparathyroidism, deafness and renal dysplasia syndrome.
Literature	GENERIF_SUMMARY	There is an association between expression of Th1/Th2 transcription factors and cytokines (T-bet, GATA-3, IFN-gamma, IL-4,IL-18) in systemic lupus erythematosus.
Literature	GENERIF_SUMMARY	No mutations were identified in patients with isolated hypoparathyroidism, thereby indicating that GATA3 abnormalities are more likely to result in two or more of the phenotypic features of the HDR syndrome.
Literature	GENERIF_SUMMARY	The human GATA-3 gene is not regulated in response to polarizing signals that are sufficient to direct Th2-specific expression in mouse cells.
Literature	GENERIF_SUMMARY	E2A proteins prevent lymphoma cell expansion, at least in part through regulation of Gfi1b and modulation of Gata3 expression.
Literature	GENERIF_SUMMARY	GATA-3 nuclear translocation is dependent on its phosphorylation on serine residues by p38 MAPK
Literature	GENERIF_SUMMARY	GATA3 establishes a positive feedback loop that increases transgenic T-cell receptor surface expression in developing CD4 lineage cells.
Literature	GENERIF_SUMMARY	Gata-3 has a role in specifying and maintaining mammary cell fate
Literature	GENERIF_SUMMARY	WNT8B expression in hepatocyte progenitors derived from human ES cells is due to POU5F1 (OCT3/OCT4) and GATA3, and WNT8B expression in diffuse-type gastric cancer is due to POU5F1 and GATA6
Literature	GENERIF_SUMMARY	In Dermatophagoides farinae-stimulated PBMCs from patients with asthma, expression of GATA-3 and T-bet and FOPX3 expression is decreased.
Literature	GENERIF_SUMMARY	In patients with acute asthma, rosiglitazone could regulate the balance of IFN-gamma and IL-4 by affecting expression of T-bet mRNA and GATA-3.
Literature	GENERIF_SUMMARY	GATA-3 is a critical component of the master cell-type-specific transcriptional network including ER alpha and FoxA1 that dictates the phenotype of hormone-dependent breast cancer
Literature	GENERIF_SUMMARY	Activation of GATA-3 might be one of the mechanisms for induction of IL-5 expression in chronic rhinosinusitis.
Literature	GENERIF_SUMMARY	GATA-3 expression in bone marrow stromal cells from

Main Category	Category	Terms
		chronic aplastic anemia was significantly higher than controls. Expression levels of GATA genes may influence hematopoiesis in BM microenvironment & relate to the pathogenesis/development of AA.
Literature	GENERIF_SUMMARY	Gata3 is a critical element determining inductive T helper (Th) type 2 cell differentiation and limiting Th1 differentiation by Notch.
Literature	GENERIF_SUMMARY	Transactivation by GATA-3 increased luciferase expression 100-fold, in the absence of stimulation. Synergistic interactions were demonstrated between Ets1, GATA-3 and AP-1.
Literature	GENERIF_SUMMARY	Gene-gene interaction between GATA3 and IL13 polymorphisms can influence the risk of childhood rhinitis.
Literature	GENERIF_SUMMARY	Direct interactions between NKX2-5 and GATA3 as indicated by co-immunoprecipitation data may contribute to MEF2C regulation.
Literature	GENERIF_SUMMARY	Role in the proliferaative and regenerative capacity of mammary stem and progenitor cells.
Literature	GENERIF_SUMMARY	The expression patterns of T-Bet and GATA-3 oppose progesterone receptor, suggesting antagonistic function and/or regulation between PR and T-Bet/ GATA-3
Literature	GENERIF_SUMMARY	The expression of GATA-3 was negatively correlated to the expression of IL-12 in patients with allergic rhinitis.
Literature	GENERIF_SUMMARY	GATA-3 is a breast cancer marker almost exclusively expresses among ER-positive tumors.
Literature	GENERIF_SUMMARY	A SNP in GATA3 is associated with atopic eczema. This finding highlights the importance of GATA3 as an immune-modulating gene in atopic eczema.
Literature	GENERIF_SUMMARY	GATA3 is hypothesized to be integral to the ERalpha pathway
Literature	GENERIF_SUMMARY	We aimed to evaluate the association between the SNPs of GATA-3 and AR.
Literature	GENERIF_SUMMARY	GATA-3 mRNA expression level was significantly higher preoperatively, and remained higher by postoperative d 7 in patients with postoperative infection after aggressive hepatobiliary pancreatic surgery.
Literature	GENERIF_SUMMARY	the T cell receptor-mediated induction of Gfi1 controls Th2 cell differentiation through the regulation of GATA3 protein stability
Literature	GENERIF_SUMMARY	cyclic AMP signals enhance histone H3 acetylation at the IL-5 promoter and the concerted binding of GATA-3 and NFATc to the promoter.
Literature	GENERIF_SUMMARY	The SNP at rs1269486 of GATA3 is associated with allergic rhinitis in Chinese.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	GlcNAc6ST-1 transcription is coordinated with the NF-kappaB/GATA-3 axis, which is known to figure heavily in Th2 cell differentiation
Literature	GENERIF_SUMMARY	Data demonstrate that zinc influences the proliferation and differentiation, and increases GATA-3 expression, of CD34(+) progenitors both in young and old ages.
Literature	GENERIF_SUMMARY	A novel mutation in the GATA3 gene in a patient with PTH-deficient hypoparathyroidism.
Literature	GENERIF_SUMMARY	reviews the literature from the past 10 y on GATA-3 in normal and pathological states of the mammary gland [review]
Literature	GENERIF_SUMMARY	there is no evidence that either GATA3 rs570613, or any variant in strong linkage disequilibrium with it, is associated with breast cancer risk in women.
Literature	GENERIF_SUMMARY	no statistically significant differences in recurrence or survival rates between GATA3-positive and GATA3-negative breast carcinomas
Literature	GENERIF_SUMMARY	expression of S100P, GATA3, and p63 by a majority of ovarian Brenner tumors underscores the similarity between these neoplasms and urothelial proliferations of bladder origin.
Literature	GENERIF_SUMMARY	It is possible that GATA3 mutations occur earlier in the evolution of breast tumors, compared to BRCA1, BRCA2 or sporadic tumors, and are therefore easier to detect by direct sequencing in the presence of some stromal contamination.
Literature	GENERIF_SUMMARY	GATA3 was found to bind the site located at -954/-855 and to be a key regulator of abundant KLK1 expression in keratinocyte.
Literature	GENERIF_SUMMARY	GATA-3 expression protects against severe joint inflammation and bone erosion in mice, accompanied by reduced differentiation of Th17 cells, but not Th1 cells, during mBSA-induced arthritis.
Literature	GENERIF_SUMMARY	heterozygous nonsense mutation p.R367X of the GATA3 gene in a Chinese family with the HDR syndrome
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	decreased levels in kidney and PBMC of patients with mesangial proliferative nephritis
Literature	GENERIF_SUMMARY	it might be speculated that genetic variations in GATA3 have no major influence on the origin of asthma.
Literature	GENERIF_SUMMARY	acquisition of IL-4Ralpha expression after stimulation was inhibited by IFN-lambda1, as was GATA3 expression
Literature	GENERIF_SUMMARY	p18(INK4C) is a downstream target of GATA3, constrains luminal progenitor cell expansion, and suppresses luminal tumorigenesis in the mammary

Main Category	Category	Terms
		gland
Literature	GENERIF_SUMMARY	The overexpression of GATA-3 in patients with allergic rhinitis was related to the increase of ICAM-1.
Literature	GENERIF_SUMMARY	GATA-3 is upregulated in peripheral blood mononuclear cells from patients with minimal change nephrotic syndrome.
Literature	GENERIF_SUMMARY	GATA3 inhibits primary breast tumor outgrowth and reduces lung metastatic burden by regulating key genes involved in metastatic breast tumor progression.
Literature	GENERIF_SUMMARY	FOXA1 and GATA-3 have roles in breast cancer, but FOXA1 is expressed in patients with good outcome, while GATA-3 is a luminal marker
Literature	GENERIF_SUMMARY	GATA3 binds to regulatory elements and controls target gene expression through physical interaction with core promoter regions.
Literature	GENERIF_SUMMARY	T-bet and GATA-3 play a part role in the regulation of Th1/Th2 balance in esophageal cancer.
Literature	GENERIF_SUMMARY	upregulated expression in nasal polyps tissue in Chinese patients
Literature	GENERIF_SUMMARY	GATA3 is indispensable for Ang-1-Tie2-mediated signaling in large vessel endothelial cells
Literature	GENERIF_SUMMARY	From the described expression pattern, it is highly probable that GATA-3 plays a role in follicular and epidermal morphogenesis.
Literature	GENERIF_SUMMARY	A novel missense HDR-associated GATA3 mutation, Thr272Ile, has been identified and shown to result in reduced DNA binding, a partial loss of FOG2 interaction, and a decrease in gene transcription.
Literature	GENERIF_SUMMARY	HuR binds to the GATA3 3'UTR. Inhibition of HuR decreased GATA3 mRNA, mRNA stability & protein expression.
Literature	GENERIF_SUMMARY	GATA3 binds to the promoter regions of genes responding to low ionizing radiation doses, and that silencing this protein in irradiated human keratinocytes leads to generalised transcriptional deregulation after X-irradiation.
Literature	GENERIF_SUMMARY	This review highlights recent understanding of GATA3 in both normal mammary development and tumor differentiation.
Literature	GENERIF_SUMMARY	GATA-3 occupies genes in both Th1 and Th2 cells and, unexpectedly, shares a large proportion of targets with T-bet.
Literature	GENERIF_SUMMARY	Compared with the normal controls, T-bet mRNA expression decreased whereas GATA-3 mRNA expression increased significantly in patients with systemic lupus erythematosus.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This study evaluates the association of Forkhead-box protein A1 (FOXA1) and GATA-binding protein 3 (GATA3) expressions with Oncotype DX recurrences scores in 77 cases of patients with ER-positive node-negative breast carcinomas
Literature	GENERIF_SUMMARY	I3C is a novel anticancer agent in human cancers that coexpress ERalpha, GATA3, and AhR, a combination found in a large percentage of breast cancers but not in other critical ERalpha target tissues essential to patient health
Literature	GENERIF_SUMMARY	GATA3 as a gene involved in transcriptional regulation of Cyclin D1.
Literature	GENERIF_SUMMARY	GATA3 drives invasive breast cancer cells to undergo the reversal of epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	Overexpression of GATA-3 restores interleukin (IL)-4 expression in Src family kinase Lck-deficient T helper (Th) type 2 cells; this indicates that decreased interleukin (IL)-4 expression is due in part to reduced amounts of GATA-3.
Literature	GENERIF_SUMMARY	lower expression in PBMC of psoriatic patients
Literature	GENERIF_SUMMARY	GATA3-NFAT1 complex may play an important role in the regulation of IL-13 transcription in human T cells.
Literature	GENERIF_SUMMARY	Down-regulation of GATA3 in Hodgkin lymphoma cell lines demonstrated its role in the regulation of IL-5, IL-13, STAT4, and other genes.
Literature	GENERIF_SUMMARY	Translational increase via T cell receptor (TCR) signaling is specific for GATA-3 and responsible for the TCR-dependent increase in GATA-3 associated with enhanced T helper type (Th)2 cell differentiation in T cells from TCR transgenic mice.
Literature	GENERIF_SUMMARY	Three new susceptibility loci at 2p16.1 (rs1432295, REL, 8q24.21 (rs2019960, PVT1 and 10p14 (rs501764, GATA3).
Literature	GENERIF_SUMMARY	The T-bet deficiency and the correlation of GATA-3 expression with the increase of IgE are characteristics of long-lasting asthma.
Literature	GENERIF_SUMMARY	The Cys321Ser-mutated GATA3 lacked transactivation activity. Moreover, mutated GATA3 exerted a dominant-negative effect over the transactivation activity of wild-type GATA3.
Literature	GENERIF_SUMMARY	report clinical phenotype and molecular analysis of GATA3 in 5 Japanese HDR patients, including 2 familial cases; sequence analysis of GATA3 demonstrated three novel (R262G, c1063delC and C318) and two reported mutations (c.432insG and c.1051-1G>T)
Literature	GENERIF_SUMMARY	GATA3 abnormalities in patients with HDR syndrome; results consistent with previous finding that GATA3 mutations are usually identified in patients with 2 or 3 of the HDR triad features

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	predominance of intraglomerular T-bet expression relative to GATA3 expression associated with poor response to treatment with bolus steroid
Literature	GENERIF_SUMMARY	GATA3 may play a crucial role in oestrogen receptor positive breast cancer patients
Literature	GENERIF_SUMMARY	Plasmacytoid carcinoma of the bladder is a tumor entity, characterized by specific immunohistochemical markers including positivity for GATA-3.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	The ratio of T-bet/GATA-3 can be used as an objective indicator of immune imbalance in patients with allergic asthma.
Literature	GENERIF_SUMMARY	The prototypic Th2 cytokine IL-4 was the only cytokine capable of inducing GATA3 in skin explants from healthy donors.
Literature	GENERIF_SUMMARY	CD8 T cells express significantly higher levels of GATA-3 and that this increase is associated with augmented production of IL-13 and is more pronounced in patients with diffuse cutaneous systemic sclerosis.
Literature	GENERIF_SUMMARY	Allergic inflammatory responses are orchestrated by several transcription factors, particularly NF-kappaB and GATA3. (Review)
Literature	GENERIF_SUMMARY	CDKN2A, GATA3, CREBBP, ITGA2, NBL1 and TGM4 were down-regulated in the prostate carcinoma glands compared to the corresponding normal glands
Literature	GENERIF_SUMMARY	These results demonstrate a direct positive role of estrogen in regulating Aurora-A expression through activation of the ERalpha-GATA-3 signaling cascade
Literature	GENERIF_SUMMARY	All the enhanceosome components comprising estrogen receptor alpha (ERalpha), FOXA1, and GATA3 are necessary for the full repertoire of cancer-associated effects of the ERalpha.
Literature	GENERIF_SUMMARY	Expression of GATA3 reprograms BTNBCs to a less aggressive phenotype and inhibits a major mechanism of metastasis through inhibition of LOX.
Literature	GENERIF_SUMMARY	A strong association was evident between GATA-3 and caspase-14 expression in preinvasive ductal carcinoma in situ samples, where GATA-3 also displayed prognostic significance
Literature	GENERIF_SUMMARY	Compared with transplant recipients not exhibiting operational tolerance operationally tolerant individuals had a predominant regulatory gene expression profile with higher expression of GATA3, TGFB1, TGFBR1, and FOXP3 which was stable over time.
Literature	GENERIF_SUMMARY	c-Myb critically regulates IL-13 expression via direct binding to the conserved GATA-3 response element

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the BRCA1-GATA3 interaction is important for the repression of genes associated with triple-negative and basal-like breast cancer (BLBCs) including FOXC1
Literature	GENERIF_SUMMARY	Leukocyte T-box expressed in T cells (TBET)/GATA3 mRNA ratio is elevated in patients with acute coronary syndrome, and highest in patients with acute myocardial infarction.
Literature	GENERIF_SUMMARY	Th1/Th2 imbalance exists in the acute phase of childhood Hench-Schonlein purpura, which correlates with the abnormal expression of transcription factor T-bet and GATA-3.
Literature	GENERIF_SUMMARY	GATA binding protein 3 is down-regulated in bladder cancer yet strong expression is an independent predictor of poor prognosis in invasive tumor.
Literature	GENERIF_SUMMARY	GATA3 is a sensitive and specific marker for the diagnosis of breast and urothelial carcinomas.
Literature	GENERIF_SUMMARY	Dysregulation of Th1 transcription may contribute to heightened expression of STAT6 and GATA3 leading to exaggerated Th2-driven manifestations of allergic disease.
Literature	GENERIF_SUMMARY	GATA3 tethers TCF7L2 to the genome at a subset of sites in MCF7 cells.
Literature	GENERIF_SUMMARY	this study demostrtrates a new T-cell regulatory pathway that directly links decreased proteasome degradation of GATA3, CTLA-4 upregulation, and inhibition of T-cell responses.
Literature	GENERIF_SUMMARY	GATA3 IHC is a sensitive marker for uro-thelial carcinoma and positive staining is typically nonfocal and moderate or strong in intensity.
Literature	GENERIF_SUMMARY	GATA3 enhance induction of POU4F3 and myosin VII
Literature	GENERIF_SUMMARY	report that the transcription factor GATA3 was highly expressed by human type 2 innate lymphoid cells
Literature	GENERIF_SUMMARY	GATA3 is pivotal in mediating enhancer accessibility at regulatory regions involved in ESR1-mediated transcription
Literature	GENERIF_SUMMARY	This report adds to the body of data on genotype-phenotype analysis in HDR syndrome.
Literature	GENERIF_SUMMARY	There appears to be no link between GATA-3 positivity and survival in men with breast cancer, whereas in women, GATA-3-positive tumors are typically lower grade with a better prognosis.
Literature	GENERIF_SUMMARY	Brain calcification caused by HDR syndrome with GATA3 mutation, presenting as a familial brain calcinosis.
Literature	GENERIF_SUMMARY	Gene expression of T-bet and GATA-3 by endometrial stromal cells is controlled by hormonal conditions mimicking decidualization, and cyclic AMP.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	GATA3/miR-29b axis regulates the breast tumor tumour microenvironment and inhibits metastasis.
Literature	GENERIF_SUMMARY	GATA3 may be a useful marker for metastatic breast carcinoma, especially triple-negative and metaplastic carcinomas its labeling may help distinguish metaplastic carcinoma from malignant phyllodes tumors.
Literature	GENERIF_SUMMARY	Identification of the E3 deubiquitinase ubiquitin-specific peptidase 21 (USP21) as a positive regulator of the transcription factor GATA3
Literature	GENERIF_SUMMARY	Data indicate that T-bet mRNA and the ratio of T-bet
Literature	GENERIF_SUMMARY	Majority of human prostate tumors show loss of active GATA3 as they progress to the aggressive castrate-resistant stage.
Literature	GENERIF_SUMMARY	Iranian women who carry the GATA3 gene's 17-CT dinucleotide repeat are at lower risk of developing breast cancer.
Literature	GENERIF_SUMMARY	Thus, approximately 20 % ER-positive breast cancers have somatic GATA3 mutations that lead to a loss of GATA3 transactivation activity and altered cell invasiveness
Literature	GENERIF_SUMMARY	Increase in the expression of GATA-3 is associated with endoscopic inflammation in patients with ulcerative colitis.
Literature	GENERIF_SUMMARY	GATA3 lacks sensitivity for distinguishing sarcomatoid renal cell carcinoma from atypical epithelioid epithelioid angiomyolipomas
Literature	GENERIF_SUMMARY	Data established a seven-gene (AR, ESR2, GATA3, GBX2, KRT16, MMP28 and WNT11) prognostic signature to define a subset of triple-negative breast cancer (TNBC).
Literature	GENERIF_SUMMARY	GATA3 overexpression results in a reduction of TGFss1 response in MDA-MB-231 (MB-231) cell line
Literature	GENERIF_SUMMARY	we evaluated GATA3 expression in urinary bladder paragangliomas, which may closely mimic urothelial carcinomas. GATA3 was positive in 83% of urinary bladder paragangliomas. this will prevent misdiagnosis of urothelial tumor based on GATA3 expression
Literature	GENERIF_SUMMARY	GATA3 immunohistochemistry is not helpful in resolving the differential diagnosis between a primary salivary gland neoplasm and metastatic breast cancer
Literature	GENERIF_SUMMARY	Report utility of gata3 as an immunohistochemical marker for tumors of the autonomic nervous system.
Literature	GENERIF_SUMMARY	our present study clarifies that the aberrant expression profile of GATA-3 in human clear cell renal carcinoma is possibly involved with tumorigenesis, and the complicated mechanism is worthy of further investigation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Review
Literature	GENERIF_SUMMARY	The expression of GATA3 and TTF1 distinguishes metastatic mammary carcinoma from primary lung carcinoma.
Literature	GENERIF_SUMMARY	Variation at 10p12.2 (PIP4K2A) and 10p14 (GATA3) influences risk of acute lymphoblastic anemia and tumor subtype. (Meta-analysis)
Literature	GENERIF_SUMMARY	GATA-3 is rarely positive in bladder adenocarcinomas displaying extracellular mucin production.
Literature	GENERIF_SUMMARY	findings demonstrate high sensitivity and specificity of the GATA3 diagnostic marker, in regional metastases of urothelial carcinoma
Literature	GENERIF_SUMMARY	TGFBR3 co-downregulated with GATA3 is associated with methylation of the GATA3 gene in bladder urothelial carcinoma.
Literature	GENERIF_SUMMARY	our study suggests that GATA3 and human papillomavirus detection are useful markers for distinguishing urothelial carcinomas from squamous cell carcinomas of the penile urethra.
Literature	GENERIF_SUMMARY	Genotype at the GATA3 SNP was also associated with early treatment response and risk of acute lymphoblastic leukemia relapse.
Literature	GENERIF_SUMMARY	GATA3 is a useful marker in the characterization of not only mammary and urothelial but also renal and germ cell tumors, mesotheliomas, and paragangliomas.
Literature	GENERIF_SUMMARY	GATA3 abrogated the canonical TGFbeta-Smad signaling by abolishing interactions between Smad4 and its DNA binding elements
Literature	GENERIF_SUMMARY	intraglomerular predominance of T-bet over GATA3 might be used as diagnosis maker of ABMR in addition to C4d, especially in C4d-negative cases.
Literature	GENERIF_SUMMARY	GATA3 is expressed in a minority of oncocytomas. We found an overall lack of GATA3 expression in primary and metastatic renal cell carcinomas.
Literature	GENERIF_SUMMARY	Mafb is responsible for executing one branch of the SGN differentiation program orchestrated by the Gata3 transcriptional network
Literature	GENERIF_SUMMARY	85-fold of that measured in untreated cells, whereas T-bet transcripts were lowered merely to
Literature	GENERIF_SUMMARY	inhibits proliferation and induces differentiation of primary keratinocytes
Literature	GENERIF_SUMMARY	Inhibition of GATA3 activity or blockade of GATA3 expression may attenuate the interleukin-13 mediated asthma phenotypes.
Literature	GENERIF_SUMMARY	GATA-3 expression is associated with a less aggressive phenotype and a better prognosis in patients with

Main Category	Category	Terms
		hormone receptor-positive/HER2-negative breast cancer.
Literature	GENERIF_SUMMARY	GATA3 may be a useful addition to immunostaining panels for serous effusion specimens when metastatic breast carcinoma is a consideration.
Literature	GENERIF_SUMMARY	Dutch family with a new mutation (c523_528dup) in GATA3 causing HDR syndrome. Their hearing impairment was congenital, bilateral and symmetric. Audiograms showed mild-to-moderate hearing impairment with a flat audiogram configuration.
Literature	GENERIF_SUMMARY	GATA3 plays an important role in the prevention of bladder cancer progression and metastasis by inhibiting cell migration and invasion as well as epithelial-to-mesenchymal transition.
Literature	GENERIF_SUMMARY	GATA3 mutations mainly occur in patients with luminal-like breast cancer and have identifiable clinicopathologic and genetic characteristics, highlighting a subgroup of patients with breast cancer in whom limited therapy may be appropriate.
Literature	GENERIF_SUMMARY	GATA3 expression independently predicts an unfavorable prognosis in primary gastric adenocarcinoma patients.
Literature	GENERIF_SUMMARY	GATA3 immunohistochemistry is a valuable addition in the discrimination of cystic or oncocytic parathyroid lesions
Literature	GENERIF_SUMMARY	High expression of FOXA1 and GATA3 was significantly associated with improved disease-free survival.
Literature	GENERIF_SUMMARY	Significant association with alleles of two STAT4 markers and nominal association of Autoimmune Addison's disease with alleles at GATA3, is reported.
Literature	GENERIF_SUMMARY	Patients with GATA3 mutation present with early-onset sensorineural hearing loss
Literature	GENERIF_SUMMARY	Abnormal expression of the transcription factors T-bet and GATA-3 contributes to the imbalance of Thl/Th2 lymphocytes associated with immune dysfunction, leading to the development and progression of aplastic anemia.
Literature	GENERIF_SUMMARY	High GATA3 expression is associated with metastatic urothelial carcinoma and distinguishes between metastatic urothelial carcinoma and squamous cell carcinomas.
Literature	GENERIF_SUMMARY	GATA-3 is, so far, the best breast-specific immunomarker, especially when encountering ER-negative metastatic breast carcinomas
Literature	GENERIF_SUMMARY	GATA-3 expression differed among urothelial carcinoma variants. GATA-3 is a useful marker for confirming the urothelial origin of micropapillary and plasmacytoid urothelial carcinoma variants but not that of sarcomatoid or small cell carcinoma variants.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	with a similar distribution across the genome, comparing to T47D cells. CONCLUSIONS
Literature	GENERIF_SUMMARY	Study shows the capacity of human CD4(+) and CD8(+) T cells for stable co-expression of GATA-3 and T-bet in humanized mice and reveals a critical role for IL-12 in regulating this phenotype.
Literature	GENERIF_SUMMARY	IFN-alpha/beta disrupts the GATA3-autoactivation loop and promotes epigenetic silencing of a Th2-specific regulatory region within the GATA3 gene.
Literature	GENERIF_SUMMARY	Data propose that control of GATA3 levels by Fbw7 contributes to the fine-tuning of T-cell development.
Literature	GENERIF_SUMMARY	GATA3 is expressed in most cases of sarcomatoid urothelial carcinoma.
Literature	GENERIF_SUMMARY	The effect of GATA3 on SPINK5 expression was indirect and GATA3 alone was insufficient for final differentiation of keratinocytes where full SPINK5 expression was observed.
Literature	GENERIF_SUMMARY	Our novel finding of GATA3 positivity in one-third of bladder small cell carcinoma is of potential value in differentiating small cell carcinomas of prostate origin from those of bladder origin.
Literature	GENERIF_SUMMARY	Strong GATA-3 expression is a reliable marker of primary breast carcinoma in the appropriate clinical context. GATA-3 reactivity in around 70% of triple-negative breast carcinomas is also clinically useful.
Literature	GENERIF_SUMMARY	Identified a novel nonsense mutation which will expand the spectrum of HDR-associated GATA3 mutations.
Literature	GENERIF_SUMMARY	GATA3 is a diagnostically useful marker for triple-negative breast cancer and is more sensitive than MGB and GCDFP15 combined.
Literature	GENERIF_SUMMARY	This study expands the spectrum of neoplasms known to express GATA-3.
Literature	GENERIF_SUMMARY	ASH2L enhances ERalpha expression as a coactivator of GATA3 in breast cancers
Literature	GENERIF_SUMMARY	Data show that the expression levels of transcription factors GATA-3 and FOXP3 were upregulated with 1.0 mug/ml galectin-1, while transcription factors TBX21 and RORC expression levels were reduced with both 1.0 and 2.0 mug/ml concentrations of galectin-1.
Literature	GENERIF_SUMMARY	Highlevel GATA3 expression is associated with increased selfrenewal and proliferation of neuroblastoma cells.
Literature	GENERIF_SUMMARY	In breast cancer, some GATA3 effects shift from tumor suppressing to tumor promoting during tumorigenesis, with deregulation of three genes, BCL2, DACH1, THSD4, representing major GATA3-controlled processes in cancer progression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The commonly used breast carcinoma biomarkers vary in their prognostic implications. GCDFP-15 independently indicated a favourable prognosis. GATA-3 and MGB were not associated with outcome.
Literature	GENERIF_SUMMARY	The mRNA level of Th2-specific transcription factor GATA-3 was increased in patients with liver cancer.
Literature	GENERIF_SUMMARY	Expressions of GATA-3 in both Th2 polarized gammadeltaT cells and CD4T cells were up-regulated.
Literature	GENERIF_SUMMARY	UPII, GATA3, and p40, when used in combination, are highly sensitive in the differential diagnosis of invasive urothelial carcinoma.
Literature	GENERIF_SUMMARY	overexpression of S100A7 in A431 skin squamous carcinoma cells significantly promoted cell proliferation in vitro and tumor growth in vivo, whereas it suppressed the expression of GATA-3 and caspase-14
Literature	GENERIF_SUMMARY	Western blotting demonstrated GATA3 in choriocarcinoma, whereas the placenta, and cervical and endometrial cancer cell lines, were negative
Literature	GENERIF_SUMMARY	Identification of a novel GATA3 mutation in a deaf Taiwanese family by massively parallel sequencing
Literature	GENERIF_SUMMARY	results support that GATA binding protein 3 immunostain can be a useful tool in differentiating mesonephric lesions from endocervical and endometrial adenocarcinomas
Literature	GENERIF_SUMMARY	rs73635312[A] is associated with susceptibility to cutaneous basal cell carcinoma.
Literature	GENERIF_SUMMARY	The tested 25 SNPs in TBX21, GATA3, Rorc and Foxp3 did not associate with BD and VKH syndrome.
Literature	GENERIF_SUMMARY	GATA-3 augmentation down-regulates Connexin43 in Helicobacter pylori associated gastric carcinogenesis
Literature	GENERIF_SUMMARY	novel marker GATA3 stains a significantly higher proportion of both primary and metastatic breast carcinomas than GCDFP15 or mammaglobin with stronger and more diffuse staining, helpful in cases with small tissue samples
Literature	GENERIF_SUMMARY	Gata3 interacted with Gcm2 and MafB, two known transcriptional regulators of parathyroid development, and synergistically stimulated the PTH promoter.
Literature	GENERIF_SUMMARY	GATA3/G9A/NuRD(MTA3) complex inhibits the invasive potential of breast cancer cells in vitro and suppresses breast cancer metastasis in vivo.
Literature	GENERIF_SUMMARY	GATA3 is a highly sensitive and specific marker for mesonephric lesions in the lower genital tract; however, its utility in the upper genital tract may be more limited.
Literature	GENERIF_SUMMARY	GATA3 could be valuable in determining breast origin and in excluding a gynecologic primary.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	trefoil factor 1 demonstrated breast specificity but was inferior to mammaglobin and GATA protein binding 3
Literature	GENERIF_SUMMARY	As one of the most frequently mutated genes in breast cancer, GATA3 mutations may have an effect on DNA-binding ability, protein production, and transactivation activity
Literature	GENERIF_SUMMARY	monoclonal antibodies to GATA-3 are very sensitive reagents for the identification of breast CA, surpassing antibodies to GCDFP-15 and mammaglobin A, and offer a significant improvement in identifying TNBCs
Literature	GENERIF_SUMMARY	GATA3 expression levels in luminal breast tumors of advanced stage were significantly higher than that of the HER2 subtype and triple-negative carcinomas, as expected, but were similar to those of the luminal controls.
Literature	GENERIF_SUMMARY	Loss-of-function mutation of GATA3 is associated with Hypoparathyroidism, Deafness, and Renal Dysplasia syndrome.
Literature	GENERIF_SUMMARY	Immunohistochemical detectionof GATA-3 and FOXA1 expression is useful to differentiate breast carcinoma from other carcinomas.
Literature	GENERIF_SUMMARY	findings demonstrate that GATA3 is expressed in a wide variety of benign and malignant cutaneous epithelial neoplasms
Literature	GENERIF_SUMMARY	Defective sirtuin-1 was found to increase IL-4 expression through acetylation of GATA-3 in patients with severe asthma compared with healthy controls.
Literature	GENERIF_SUMMARY	discuss mutation of GATA3 in breast cancer and the potential mechanism(s) by which mutation may lead to a growth advantage in cancer
Literature	GENERIF_SUMMARY	Combined marker analysis of GATA3/fascin showed an inverse association and improved prognostic information for patients with breast cancer.
Literature	GENERIF_SUMMARY	GATA3 labeling is highest in estrogen receptor-positive carcinomas, it also labels estrogen receptor-negative carcinomas and thus has particular diagnostic utility in the setting of triple-negative breast carcinomas
Literature	GENERIF_SUMMARY	we confirmed that the rs10905284 is a locus associated to leprosy and influences the levels of this transcription factor in the Brazilian population.
Literature	GENERIF_SUMMARY	GATA-3 was induced by estrogen and may promote the occurrence and development of endometriosis by regulating the secretion of cytokines in the eutopic endometrial cells of endometriosis patients.
Literature	GENERIF_SUMMARY	GATA3 knockdown elevated HtrA4 expression in BeWo and JEG-3 trophoblast cell lines and enhanced the invasion activities of both lines. This study uncovered a new GATA3 function in placenta as a negative regulator of GCM1 activity and trophoblastic invasion.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These data provide mechanistic insights into GATA3-mediated chromatin reprogramming during MET, and suggest unexpected complexity to transcription factors pioneering
Literature	GENERIF_SUMMARY	this study shows that in peripheral blood, end-stage renal disease patients show increased levels of GATA-3
Literature	GENERIF_SUMMARY	GATA3 mRNA expression, but not GATA3 mutation, is an independent predictor of prolonged progression-free survival in ER-positive breast cancer patients who received first-line tamoxifen for recurrent disease.
Literature	GENERIF_SUMMARY	Dysregulation of JAM-A via p63/GATA-3 signaling pathway occurs in squamous cell carcinomas of the head and neck.
Literature	GENERIF_SUMMARY	Here, we demonstrate a novel FOXC1-driven mechanism that suppresses ERa expression in breast cancer. We find that FOXC1 competes with GATA-binding protein 3 (GATA3) for the same binding regions in the cis-regulatory elements upstream of the ERa gene and thereby downregulates ERa expression and consequently its transcriptional activity
Literature	GENERIF_SUMMARY	GATA3-mediated positive and negative feedback mechanisms via restraining Notch1 activity control human T-cell lineage commitment.
Literature	GENERIF_SUMMARY	this study highlights the molecular basis for posttranslational modifications of Gata3 that control the regulation of IFNgamma expression in memory Th2 cells.
Literature	GENERIF_SUMMARY	GATA-3 expression is frequent in triple negative breast cancer even in the absence of androgen receptors (AR). However, neither GATA-3 nor AR are associated with complete pathological response after neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	A high copy number of T-bet and GATA-3 confers susceptibility to AAU and AS, and a high copy number of FOXP3 confers susceptibility to female patients with AAU either with or without AS.
Literature	GENERIF_SUMMARY	GATA-3 is a more reliable biomarker than PTH or chromogranin A in differentiating parathyroid from thyroid lesions in cytology smears.
Literature	GENERIF_SUMMARY	C-MYC is related to GATA3 and Ki-67 expression and is associated with poor prognosis in nodal peripheral T-cell lymphomas
Literature	GENERIF_SUMMARY	GATA3 expression was correlated strongly with androgen receptor expression apocrine-type triple negative breast neoplasms.
Literature	GENERIF_SUMMARY	The aim of this study was to investigate the clinical significance of three immune cell-related transcription factors, T-bet, GATA-3 and Bcl-6 in bladder cancer in Tunisian patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results indicate that GATA3 is an independent prognostic factor and suggest that evaluation of GATA3 expression might enable more effective clinical follow-up using prognostic stratification of STS patients
Literature	GENERIF_SUMMARY	This study provides novel evidences of the mechanisms integrating Notch and TNF-alpha signaling in the transcriptional induction of GATA3 and HOXB4.
Literature	GENERIF_SUMMARY	GATA3-driven expression of miR-503 inhibits prostate cancer progression by repressing ZNF217 expression.
Literature	GENERIF_SUMMARY	MIR-720 suppresses M2 macrophage polarization by targeting GATA3; MIR-720 is down-regulated in breast cancer macrophages.
Literature	GENERIF_SUMMARY	Mutation in the GATA-3 gene is associated with hypoparathyroidism, deafness and renal dysplasia syndrome.
Literature	GENERIF_SUMMARY	Study identified the novel role of transcription factors GATA-2 and GATA-3 in suppressing MICA/B expression in HBV-infected human hepatoma cells.
Literature	GENERIF_SUMMARY	Letter/Case Reports
Literature	GENERIF_SUMMARY	this study shows that thiazolidin-4-one ring activates invariant natural killer T cells proliferation, increases the production of Th2 cytokines, and induces Th0 differentiation into Th2 subset via GATA 3 signaling
Literature	GENERIF_SUMMARY	These results suggest that the ELK3-GATA3 axis is a major pathway that promotes metastasis of breast cancer MDA-MB-231 cells
Literature	GENERIF_SUMMARY	we find an estrogen receptor-independent synthetic lethal interaction between a GATA3 frameshift mutant with an extended C-terminus and the histone methyltransferases G9A and GLP, indicating perturbed epigenetic regulation
Literature	GENERIF_SUMMARY	These results suggest that high GATA3 expression is a predictor of a poor prognosis in peripheral T cell lymphoma, and that T lymphoma cells promote M2-type macrophage differentiation through a GATA3-dependent mechanism.
Literature	GENERIF_SUMMARY	These results suggest that whereas GATA-3 and uroplakin II may be most useful in the diagnosis of urothelial carcinoma metastasis, p40 may be additionally suited as a prognostic marker.
Literature	GENERIF_SUMMARY	The GATA3 rs3824662 A allele and AA genotype may be risk factors for the development of pediatric ALL especially B-ALL in the studied cohort of Egyptian patients. The AA genotype is associated with shorter DSF, increased incidence of relapse and poor prognosis in pediatric ALL.
Literature	GENERIF_SUMMARY	We demonstrate that engagement of the T-cell receptor (TCR) on malignant T cells culminates in NF-kappaB activation and the upregulation of GATA-3 expression,

Main Category	Category	Terms
		both of which regulate the growth and survival of conventional T cells, and were shown to promote chemotherapy resistance in malignant T cells
Literature	GENERIF_SUMMARY	overexpression of GATA3 and FOXA1 cooperate with PPAR activation to drive transdifferentiation of a basal bladder cancer cells to a luminial phenotype.
Literature	GENERIF_SUMMARY	GATA3 expression is positively correlated with ER-positive, PR-positive, and HER2-positive carcinomas
Literature	GENERIF_SUMMARY	Our study has demonstrated that GATA3 expression is common in primary triple-negative breast carcinomas
Literature	GENERIF_SUMMARY	GATA3 expression, regardless of the subtype, may have a prognostic significance for breast carcinomas through its ability to promote the differentiation of luminal progenitor cells.
Literature	GENERIF_SUMMARY	We have identified ZPO2 as a negative regulator of GATA3, thus providing an alternative mechanism that results in a reduction in or perhaps even loss of GATA3 during breast cancer development.
Literature	GENERIF_SUMMARY	Our findings suggest that GATA3 stabilizes HIF-1alpha to enhance cancer invasiveness under hypoxia
Literature	GENERIF_SUMMARY	A decreased mRNA expression of GATA3 and TGF-beta was found in active VKH patients, which was correlated negatively with the DNA methylation of these factors.
Literature	GENERIF_SUMMARY	As a transcriptionally regulated program, urothelial differentiation operates as a heterarchy, wherein GATA3 is able to co-operate with FOXA1 to drive expression of luminal marker genes, but that P63 has potential to transrepress expression of the same genes.
Literature	GENERIF_SUMMARY	GATA3 is often positive in benign prostate glands with radiation atypia, which along with the morphologic features present a pitfall for the misdiagnosis of urothelial carcinoma.
Literature	GENERIF_SUMMARY	In Chinese family 7121, a heterozygous nonsense mutation c.826C>T (p.R276*) was identified in GATA3. All the three affected members suffered from sensorineural deafness and hypocalcemia; however, renal dysplasia only appeared in the youngest patient. Furthermore, an overview of thirty HDR syndrome families with corresponding GATA3 mutations revealed that hearing impairment occurred earlier in the younger generation in
Literature	GENERIF_SUMMARY	Benign prostate glands with radiation atypia show diffuse positivity for GATA3
Literature	GENERIF_SUMMARY	In resected lung adenocarcinoma, high GATA3 expression is associated with poorer prognosis for both overall survival and disease free survival.
Literature	GENERIF_SUMMARY	The coexpression of GATA3 and MYB might be helpful in the distinction of primary cutaneous adnexal carcinoma versus metastatic breast, salivary gland, or

Main Category	Category	Terms
		urothelial carcinoma
Literature	GENERIF_SUMMARY	GATA3 may be useful as a part of a panel of immunohistochemical markers in narrowing the differential diagnosis between pheochromocytoma and adrenal cortical carcinoma
Literature	GENERIF_SUMMARY	It gene expression is up-regulated by EP300-ZNF384 fusion gene product.
Literature	GENERIF_SUMMARY	GATA3 activation was diminished upon cultivation of T cells with RNase 7.
Literature	GENERIF_SUMMARY	Meta-analysis revealed that high expression of GATA3 in breast cancer is associated with improved time to tumor progression.
Literature	GENERIF_SUMMARY	Results showed that GATA3 along with TRPS1 are distinctively overexpressed in breast cancer (BC) among all GATA family members and predict better survival in patients with BC. GATA3 is then a distinctive biomarker and essential prognostic factor in BC.
Literature	GENERIF_SUMMARY	Loss of GATA3 expression was found to be an independent predictor of poor patient outcomes in bladder urothelial carcinoma.
Literature	GENERIF_SUMMARY	GATA3 expression is associated with breast carcinomas of luminal subtype and low histological grade.
Literature	GENERIF_SUMMARY	Data delineates the regulatory mechanisms of GATA3 in DNA double-strand breaks repair and strongly suggests that it might act as a tumor suppressor by promoting CtIP expression and homologous recombination to stabilize genomes.
Literature	GENERIF_SUMMARY	GATA-3 in lymphocytes showed the increased expression of GATA-3 in elderly people in comparison with the young. Women demonstrated higher expression of GATA-3 in compare with men.
Literature	GENERIF_SUMMARY	Authors demonstrated that ER(alpha), GATA3 and FOXA1 form a transcriptional complex with Ell3 to regulate IL-20 expression in ER(+) breast cancer cells. FOXA1 represses IL-20 expression, whereas GATA3 and ER(alpha) activate it.
Literature	GENERIF_SUMMARY	GATA-3 is a sensitive and specific marker for the diagnosis of acute leukemias with T-cell differentiation.
Literature	GENERIF_SUMMARY	data provide new insights into the role of SEMA3B in mammary gland and provides a new branch of GATA3 signaling that is pivotal for inhibition of breast cancer progression and metastasis
Literature	GENERIF_SUMMARY	This study demonstrates consistent GATA-3 staining in walthard nests, rare staining in adenomatoid tumors, and is otherwise rarely positive in normal urologic and gynecologic mesothelia. Furthermore, GATA-3 is uniformly positive in epididymi and negative in efferent ductules which may reflect the embryological

Main Category	Category	Terms
		evolvement of these tissues.
Literature	GENERIF_SUMMARY	In summary, we documented a transcriptional regulation of GATA3 in glioblastoma cells upon transfection with GAB sequence.
Literature	GENERIF_SUMMARY	Rs17144046 located near GATA3 was significantly associated with benign prostatic hyperplasia and lower urinary tract symptoms.
Literature	GENERIF_SUMMARY	GATA3 is a sensitive marker for primary genital extramammary paget disease
Literature	GENERIF_SUMMARY	Retrospective immunohistochemical analysis of GATA3 expression in 164 breast cancer metastases showed a striking difference between mammaglobin and GATA3 expression (51.2% vs 94% positivity). These findings highlight GATA3 as a more reliable and sensitive diagnostic marker for breast cancer metastases and possibly metastatic tumors of unknown origin than mammaglobin.
Literature	GENERIF_SUMMARY	Coexpression of GATA3 and CK7 in most clear cell papillary renal cell carcinomas provides evidence of their origin from distal nephron and can be used in differential diagnosis.
Literature	GENERIF_SUMMARY	FBXW7alpha inhibits breast cancer cells survival by promoting GATA3 degradation/destabilizing GATA3.
Literature	GENERIF_SUMMARY	KMT1A positively regulated the self-renewal and tumorigenicity of human Bladder cancer stem cells via KMT1A-GATA3-STAT3 circuit, in which KMT1A could be a promising target for bladder cancer therapy.
Literature	GENERIF_SUMMARY	GATA3 Immunohistochemical Staining in Hodgkin Lymphoma
Literature	GENERIF_SUMMARY	results demonstrate that GATA3 is involved in the regulation of filaggrin and filaggrin-2 expression during inflammatory conditions in the skin.
Literature	GENERIF_SUMMARY	The data have demonstrated that GATA3 is by far the most reliable breast-specific immunomarker in both surgical and cytological specimens
Literature	GENERIF_SUMMARY	GATA3 is a reliable diagnostic marker for Neuroblastomas not only in scant/limited surgical specimens but also in cytologic samples, including air-dried touch imprints, which have previously been undescribed for this marker.
Literature	GENERIF_SUMMARY	the model of mutually antagonistic differentiation programs driven by mutually exclusively expressed T-bet or GATA-3 does not completely explain natural CD4 T cell priming outcomes
Literature	GENERIF_SUMMARY	Authors found a high sensitivity for all the markers analyzed. Moreover, the expression of NY-BR-1 and GATA-3 seemed the most effective for labelling male breast cancer in primary and metastatic setting.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	GATA 3 is more sensitive marker than mammaglobin and GCDFP-15 for diagnosing metastatic breast carcinoma in cytological cell block materials
Literature	GENERIF_SUMMARY	GATA-3 expression was the highest in luminal breast carcinomas, and showed higher sensitivity than GCDFP-15 and MGB. However, in the poorly differentiated IBC, its utility was still limited. One should be aware of the possible GATA-3 expression in lung carcinomas.
Literature	GENERIF_SUMMARY	We show that MSK1 downregulation impairs the differentiation of breast cancer cells, increasing their bone homing and growth capacities. MSK1 controls the expression of genes required for luminal cell differentiation, including the GATA3 and FOXA1 transcription factors, by modulating their promoter chromatin status.
Literature	GENERIF_SUMMARY	Increased GATA-3 and T-bet expression in eosinophilic esophagitis.
Literature	GENERIF_SUMMARY	GATA3 staining to differentiate primary cutaneous apocrine cribriform carcinoma from skin breast cancer metastasis has a high negative predictive value
Literature	GENERIF_SUMMARY	GATA3 interacts with and is acetylated by the acetyltransferase CBP. The major acetylated site of GATA3 in lung adenocarcinoma cells is lysine 119.
Literature	GENERIF_SUMMARY	one allele of the GATA3 second zinc finger leads to loss of binding and decreased expression at a subset of genes, including Progesterone Receptor
Literature	GENERIF_SUMMARY	Efficacy of GATA3 Versus BRST2 for the Identification of Metastatic Breast Carcinoma in the Upper GI Tract.
Literature	GENERIF_SUMMARY	demonstrated a correlation between GATA3 expression and only ER+ and suggest that a higher GATA3 expression is a good prognostic factor for OS for ER+ BC patients
Literature	GENERIF_SUMMARY	The data indicate that expression of CRTh2 is regulated through the competitive action of GATA3 and NFAT1.
Literature	GENERIF_SUMMARY	findings provide evidence for a new role for GATA3 in ovarian HGSC stemness, and demonstrate that GATA3 may serve as a biomarker for precision epigenetic therapy in the future
Literature	GENERIF_SUMMARY	Treg cells from asthmatic patients expressed more FOXP3 as well as GATA3; the expression level of GATA3 negatively correlated with FEV1%pred. Increased expressions of USP21 and PIM2 in Treg cells from asthmatic patients were found.
Literature	GENERIF_SUMMARY	the present study postulates that the genetic variation of the transcription factor GATA3, not STAT4, is associated with the risk of type 2 diabetes in the Bangladeshi population.
Literature	GENERIF_SUMMARY	GATA3 mutations, recently observed in breast cancer,

Main Category	Category	Terms
		encode active transcription factors.
Literature	GENERIF_SUMMARY	GATA3 was the best overall marker for mesonephric and mesonephric-like carcinomas, but cannot be used to distinguish mesonephric carcinosarcomas from Mullerian carcinosarcomas.
Literature	GENERIF_SUMMARY	Study established an important role of p38gamma MAPK in epithelial-mesenchymal transition (EMT) and identified a novel signaling pathway for p38gamma MAPK-mediated tumor promotion. p38gamma MAPK regulated miR-200b through inhibiting GATA3 by inducing its ubiquitination, leading proteasome-dependent degradation.
Literature	GENERIF_SUMMARY	GATA3 rs3824662 is not associated with susceptibility to systemic lupus erythematosus (SLE) either in adult or in pediatric patients; however, in pediatric SLE patients, the heterozygous AC genotype could be considered a risk factor for lupus nephritis (LN). At the same time, the AC and AA genotypes could be considered as predictors for LN and active renal disease.
Literature	GENERIF_SUMMARY	GATA3 and TP53INP1 were identified as targets of miR155. Exosomal miR155 inhibited these targets by directly targeting their 3' untranslated regions. Knockdown of miR1555p was observed to reverse the EMT and chemoresistant phenotypes of gastric cancer cells, potentially via GATA3 and TP53INP1 upregulation.
Literature	GENERIF_SUMMARY	A Japanese family with multiple members with hypoparathyroidism-deafness-renal dysplasia (HDR) syndrome and congenital choanal atresia was found to have a c.708_709insC (p.Ser237Glnfs*66) allelic variant in exon 3.
Literature	GENERIF_SUMMARY	GATA3-AS1 binds components of the MLL methyltransferase.
Literature	GENERIF_SUMMARY	This review summarizes the various contributions of Gata3 to haematopoiesis with a particular focus on the emergence of the first haematopoietic stem cells in the embryo-a process that appears to be influenced by Gata3 at various levels, thus highlighting the complex nature of Gata3 action.
Literature	GENERIF_SUMMARY	Although GATA3 is a more sensitive lineage marker than SOX10 in metastatic breast cancer, SOX10 is a useful adjunct because it is positive in most GATA3-negative breast metastases.
Literature	GENERIF_SUMMARY	GATA3 has an association with susceptibility to the development of moderate-to-severe asthma and is a transcription factor linked to the T-cell response in asthma and eosinophilia.
Literature	GENERIF_SUMMARY	The levels of ANP, NPRA, IL-4 and GATA3 increased in peripheral blood of children with respiratory syncytial virus bronchiolitis.
Literature	GENERIF_SUMMARY	The single nucleotide polymorphisms (SNPs)

Main Category	Category	Terms
		rs4794067, rs2275806, rs2232365, and rs3761548 map in the genes of TBX21, GATA3, and FOXP3 were investigated if they are associated with acute cellular liver transplant rejection.
Literature	GENERIF_SUMMARY	rs3824662 in GATA3 associates with CRLF2 expression in leukemic cells and predicts prognosis in pediatric BCP-ALL patients.
Literature	GENERIF_SUMMARY	Genome-wide mapping revealed a substantial co-occupancy of binding regions by ISL1 and GATA3, and ISL1 physically interacts with GATA3, and together they synergistically regulate the expression of oncogenic pathways.
Literature	GENERIF_SUMMARY	miR-374b promotes glioma process in vitro through suppressing SEMA3B via targeting GATA3
Literature	GENERIF_SUMMARY	Comprehensive analysis of the GATA transcription factor gene family in breast carcinoma using gene microarrays, online databases and integrated bioinformatics.
Literature	GENERIF_SUMMARY	Transient Expression of GATA3 in Hematopoietic Stem Cells Facilitates Helper Innate Lymphoid Cell Differentiation.
Literature	GENERIF_SUMMARY	GATA3 as a sensitive and specific marker for urothelial carcinoma can be effectively used to exclude other genitourinary malignancies, prostatic adenocarcinoma and renal cell carcinoma, at metastatic site.
Literature	GENERIF_SUMMARY	Loss of GATA3 expression was statistically associated with adverse clinicopathologic parameters, such as advanced stage, lymphovascular invasion, neural invasion, lymph node metastasis, and extensive necrosis in upper tract urothelial carcinoma.
Literature	GENERIF_SUMMARY	Incidence and Significance of GATA3 Positivity in Pancreatic Ductal Adenocarcinoma and Cholangiocarcinoma.
Literature	GENERIF_SUMMARY	CD276 and the gene signature composed of GATA3 and LGALS3 enable prognosis prediction of glioblastoma multiforme.
Literature	GENERIF_SUMMARY	Breast cancer mutation in GATA3 zinc finger 1 induces conformational changes leading to the closer binding of ZnFn2 with a wrapping architecture.
Literature	GENERIF_SUMMARY	We identified four Type 1 diabetes (T1D) risk loci reaching genome-wide significance in the Chinese Han population, including two novel loci
Literature	GENERIF_SUMMARY	This work expands the phenotypic spectrum of GATA3 mutations. It reveals the novel insight that decreased and altered GATA3 activity coincides with autoimmune arthritis. This work suggests that modulation of GATA3 may be a therapeutic approach for patients with autoimmune arthritis
Literature	GENERIF_SUMMARY	H. pylori significantly inhibited the proliferation,

Main Category	Category	Terms
		migration, and tube formation of HUVECs, as well as increased the production of the inflammatory factor CHI3L1 and phosphorylated p38 from endothelial cells along with an increased expression of GATA3.
Literature	GENERIF_SUMMARY	we identified a novel, nonredundant role for GATA3 in MCs from their first appearance in embryonic kidney to mature glomeruli in health and disease. Furthermore, GATA3 expression increases inMCs in mesangial proliferative GN and haploinsufficiency of GATA3 results in MC defects, suggesting its importance for glomerular homeostasis and response to injury.
Literature	GENERIF_SUMMARY	The expression level of GATA3 in the experimental group was significantly higher than that in the control group (p
Literature	GENERIF_SUMMARY	decrease in GATA3 staining was significantly associated with high proliferative activity and poor clinical outcome in MIBC. IHC staining for GATA3 might facilitate in risk stratification in patients with MIBC receiving radical cystectomy.
Literature	GENERIF_SUMMARY	a feedback loop between these two axes, specifically GATA3-induced miR-455-3p expression, could repress ZEB1 and its recruitment of NuRD (MTA1) to suppress miR-455, which ultimately regulates TGF-beta signaling.
Literature	GENERIF_SUMMARY	GATA3-Controlled Nucleosome Eviction Drives MYC Enhancer Activity in T-cell Development and Leukemia.
Literature	GENERIF_SUMMARY	Cooperativity of co-factor NR2F2 with Pioneer Factors GATA3, FOXA1 in promoting ERalpha function.
Literature	GENERIF_SUMMARY	GATA3 suppresses human fibroblasts-induced metastasis of clear cell renal cell carcinoma via an anti-IL6/STAT3 mechanism.
Literature	GENERIF_SUMMARY	GATA3 recruits UTX for gene transcriptional activation to suppress metastasis of breast cancer.
Literature	GENERIF_SUMMARY	In clear cell renal cell carcinoma (ccRCC) ECHS1 downregulation induced fatty acid (FA) and branched-chain amino acid (BCAA) accumulation, which inhibited AMPK-promoted expression of GATA3, a transcriptional activator of ECHS1. BCAA accumulation induced activation of mTORC1 and de novo FA synthesis and promoted cell proliferation. GATA3 expression phenocopied ECHS1 in predicting ccRCC progression and patient survival.
Literature	GENERIF_SUMMARY	This is the first study to characterize CK7 negative breast tumors in the context of clinicopathology. Profiling the CK7 negative and GATA3 negative breast cancers helps to understand the biology of these specific tumor subgroups and may aid in their diagnosis.
Literature	GENERIF_SUMMARY	GATA3 reduces the ATP level in the breast cancer microenvironment and inhibits breast cancer metastasis by up-regulating ectonucleoside triphosphate diphosphohydrolase 3 (ENTPD3).

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	P63 expression in tumor cells, even focal expression, and no GATA3 is the most helpful feature in distinguishing primary mediastinal large B-cell lymphoma from mediastinal classic Hodgkin lymphoma
Literature	GENERIF_SUMMARY	Epigenetic deregulation of GATA3 in neuroblastoma is associated with increased GATA3 protein expression and with poor outcomes.
Literature	GENERIF_SUMMARY	Effects of guluronic acid, as a new NSAID with immunomodulatory properties on IL-17, RORgammat, IL-4 and GATA-3 gene expression in rheumatoid arthritis patients.
Literature	GENERIF_SUMMARY	Our study proposed a complex KIAA1429-GATA3 regulatory model based on m6A modification and provided insights into the epi-transcriptomic dysregulation in hepatocarcinogenesis and metastasis.
Literature	GENERIF_SUMMARY	Racial Differences in the Association Between Luminal Master Regulator Gene Expression Levels and Breast Cancer Survival.
Literature	GENERIF_SUMMARY	GATA3 as a master regulator for interactions of tumor-associated macrophages with high-grade serous ovarian carcinoma.
Literature	GENERIF_SUMMARY	Utility of GATA-3 Expression in the Analysis of Pituitary Neuroendocrine Tumour (PitNET) Transcription Factors.
Literature	GENERIF_SUMMARY	Cancer-specific mutation of GATA3 disrupts the transcriptional regulatory network governed by Estrogen Receptor alpha, FOXA1 and GATA3.
Literature	GENERIF_SUMMARY	GATA3 Mediates a Fast, Irreversible Commitment to BMP4-Driven Differentiation in Human Embryonic Stem Cells.
Literature	GENERIF_SUMMARY	LOC285194 inhibits proliferation of human keratinocytes through regulating miR-616/GATA3 pathway.
Literature	GENERIF_SUMMARY	Hypoparathyroidism, deafness, and renal dysplasia syndrome
Literature	GENERIF_SUMMARY	The miR-155/GATA3/IL37 axis modulates the production of proinflammatory cytokines upon TNF-alpha stimulation to affect psoriasis development.
Literature	GENERIF_SUMMARY	LncRNA GNAS-AS1 facilitates ER+ breast cancer cells progression by promoting M2 macrophage polarization via regulating miR-433-3p/GATA3 axis.
Literature	GENERIF_SUMMARY	The GATA3 X308_Splice breast cancer mutation is a hormone context-dependent oncogenic driver.
Literature	GENERIF_SUMMARY	LncRNA GATA3-AS1 facilitates tumour progression and immune escape in triple-negative breast cancer through destabilization of GATA3 but stabilization of PD-L1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Combined SOX10 GATA3 is most sensitive in detecting primary and metastatic breast cancers
Literature	GENERIF_SUMMARY	GATA3 expression in the solid cell nest of thyroid.
Literature	GENERIF_SUMMARY	Aberrant GATA3 Staining in Prostatic Adenocarcinoma
Literature	GENERIF_SUMMARY	The potential utility of GATA binding protein 3 for diagnosis of malignant pleural mesotheliomas.
Literature	GENERIF_SUMMARY	Association of GATA3 Polymorphisms With Minimal Residual Disease and Relapse Risk in Childhood Acute Lymphoblastic Leukemia.
Literature	GENERIF_SUMMARY	Role of GATA3 exon 6 germline mutations in breast cancer progression in Egyptian female patients.
Literature	GENERIF_SUMMARY	The Relationship of GATA3 and Ki-67 With Histopathological Prognostic Parameters, Locoregional Recurrence and Disease-free Survival in Invasive Ductal Carcinoma of the Breast.
Literature	GENERIF_SUMMARY	Incidence and significance of GATA3 positivity in gallbladder adenocarcinoma.
Literature	GENERIF_SUMMARY	Role of the IL-23-T-bet/GATA3 Axis for the Pathogenesis of Ulcerative Colitis.
Literature	GENERIF_SUMMARY	GATA3 rs3824662A allele in B-cell acute lymphoblastic leukemia in adults, adolescents and young adults
Literature	GENERIF_SUMMARY	Cathepsin V suppresses GATA3 protein expression in luminal A breast cancer.
Literature	GENERIF_SUMMARY	Chromosome 2q gain and epigenetic silencing of GATA3 in microglandular adenosis of the breast.
Literature	GENERIF_SUMMARY	Nonlinear relationship between chromatin accessibility and estradiol-regulated gene expression.
Literature	GENERIF_SUMMARY	GATA3 somatic mutations are associated with clinicopathological features and expression profile in TCGA breast cancer patients.
Literature	GENERIF_SUMMARY	Utility of Immunohistochemistry for MUC4 and GATA3 to Aid in the Distinction of Pleural Sarcomatoid Mesothelioma From Pulmonary Sarcomatoid Carcinoma.
Literature	GENERIF_SUMMARY	Monoclonal/polyclonal PAX-8, PTH and GATA3 immunohistochemistry in parathyroid lesions.
Literature	GENERIF_SUMMARY	GATA3 is a useful immunohistochemical marker for distinguishing sarcomatoid malignant mesothelioma from lung sarcomatoid carcinoma and organizing pleuritis.
Literature	GENERIF_SUMMARY	Endosalpingiosis Is Negative for GATA3.
Literature	GENERIF_SUMMARY	Is immunohistochemical expression of GATA3 helpful in the differential diagnosis of transformed mycosis fungoides and primary cutaneous CD30-positive T cell

Main Category	Category	Terms
		lymphoproliferative disorders?
Literature	GENERIF_SUMMARY	Expression of uroplakin II and GATA-3 in bladder cancer mimickers
Literature	GENERIF_SUMMARY	Gata-3 and KI-67 expression in correlation with molecular subtypes of breast cancer.
Literature	GENERIF_SUMMARY	GATA3 induces mitochondrial biogenesis in primary human CD4(+) T cells during DNA damage.
Literature	GENERIF_SUMMARY	Co-expression of transcription factor AP-2beta (TFAP2B) and GATA3 in human mammary epithelial cells with intense, apicobasal immunoreactivity for CK8/18.
Literature	GENERIF_SUMMARY	Diagnostic utility of the combined use of HNF4A and GATA3 in distinction between primary and metastatic breast and gastric carcinomas.
Literature	GENERIF_SUMMARY	The estrogen receptor/GATA3/FOXA1 transcriptional network
Literature	GENERIF_SUMMARY	[Expression of GATA3 and bcl-11b in peripheral T-cell lymphoma and their clinical significance].
Literature	GENERIF_SUMMARY	ILC2 Cells Promote Th2 Cell Differentiation in AECOPD Through Activated Notch-GATA3 Signaling Pathway.
Literature	GENERIF_SUMMARY	GATA3 as a regulator for naughty cancer-associated fibroblasts in the microenvironment of high-grade serous ovarian cancer.
Literature	GENERIF_SUMMARY	Inherited GATA3 variant associated with positive minimal residual disease in childhood B-cell acute lymphoblastic leukemia via asparaginase resistance.
Literature	GENERIF_SUMMARY	Mesenchymal Stem Cell-Derived Exosome-Containing Linc00632 Regulates GATA Binding Protein-3 Expression in Allergic Rhinitis by Interacting with Enhancer of Zeste Homolog 2 to Inhibit T Helper Cell 2 Differentiation.
Literature	GENERIF_SUMMARY	Effects of Molecular Iodine/Chemotherapy in the Immune Component of Breast Cancer Tumoral Microenvironment.
Literature	GENERIF_SUMMARY	Association Between Estrogen Receptors and GATA3 in Bladder Cancer
Literature	GENERIF_SUMMARY	Mutational signatures in GATA3 transcription factor and its DNA binding domain that stimulate breast cancer and HDR syndrome.
Literature	GENERIF_SUMMARY	Alterations in T-Cell Transcription Factors and Cytokine Gene Expression in Late-Onset Alzheimer's Disease.
Literature	GENERIF_SUMMARY	[Clinical characteristics and molecular mechanisms of hypoparathyroidism related to GATA3 gene mutation].
Literature	GENERIF_SUMMARY	GATA3 positivity is associated with poor prognosis in patients with oesophageal squamous cell carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Noncoding genetic variation in GATA3 increases acute lymphoblastic leukemia risk through local and global changes in chromatin conformation.
Literature	GENERIF_SUMMARY	Diagnostic and prognostic implications of tumor expression of the GATA-3 gene in nodal peripheral T-cell lymphoma (nPTCL)
Literature	GENERIF_SUMMARY	Mammaglobin, GATA-binding protein 3 (GATA3), and epithelial growth factor receptor (EGFR) expression in different breast cancer subtypes and their clinical significance.
Literature	GENERIF_SUMMARY	TRPS1, GATA3, and SOX10 expression in triple-negative breast carcinoma.
Literature	GENERIF_SUMMARY	Clear Cell Papillary Renal Cell Tumors
Literature	GENERIF_SUMMARY	A preclinical model of peripheral T-cell lymphoma GATA3 reveals DNA damage response pathway vulnerability.
Literature	GENERIF_SUMMARY	[Expression of SOX10 and GATA3 in breast cancer and their significance].
Literature	GENERIF_SUMMARY	The Role of Polymorphisms at the Interleukin-1, Interleukin-4, GATA-3 and Cyclooxygenase-2 Genes in Non-Surgical Periodontal Therapy.
Literature	GENERIF_SUMMARY	Immunohistochemical expression of renin and GATA3 help distinguish juxtaglomerular cell tumors from renal glomus tumors.
Literature	GENERIF_SUMMARY	GATA3 Exerts Distinct Transcriptional Functions to Regulate Radiation Resistance in A549 and H1299 Cells.
Literature	GENERIF_SUMMARY	GATA3 as an immunomodulator in obesity-related metabolic dysfunction associated with fatty liver disease, insulin resistance, and type 2 diabetes.
Literature	GENERIF_SUMMARY	Long non-coding RNA MALAT1 promotes Th2 differentiation by regulating microRNA-135b-5p/GATA-3 axis in children with allergic rhinitis.
Literature	GENERIF_SUMMARY	GATA3 expression loss is linked to stage progression but is unrelated to prognosis in muscle-invasive urothelial carcinoma of the bladder.
Literature	GENERIF_SUMMARY	14-3-3tau drives estrogen receptor loss via ERalpha36 induction and GATA3 inhibition in breast cancer.
Literature	GENERIF_SUMMARY	GATA3 expression in clear cell adenocarcinoma of the lower urinary tract
Literature	GENERIF_SUMMARY	Setd2 supports GATA3[+]ST2[+] thymic-derived Treg cells and suppresses intestinal inflammation.
Literature	GENERIF_SUMMARY	GATA3 Expression in Human Tumors
Literature	GENERIF_SUMMARY	Regulation of endogenous retrovirus-derived regulatory elements by GATA2/3 and MSX2 in human trophoblast

Main Category	Category	Terms
		stem cells.
Literature	GENERIF_SUMMARY	Nucleus Pulposus Cells from Calcified Discs Promote the Degradation of the Extracellular Matrix through Upregulation of the GATA3 Expression.
Literature	GENERIF_SUMMARY	[Changes in percentage of GATA3[+] regulatory T cells and their pathogenic roles in allergic rhinitis].
Literature	GENERIF_SUMMARY	Contributions of ARID5B, IKZF1, PIP4K2A, and GATA3 Gene Polymorphisms to Childhood Acute Lymphoblastic Leukemia in a Chinese Population.
Literature	GENERIF_SUMMARY	Deletion mapping of regulatory elements for GATA3 in T cells reveals a distal enhancer involved in allergic diseases.
Literature	GENERIF_SUMMARY	ZNF503 combined with GATA3 is a prognostic factor in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	TRPS1 and GATA3 Expression in Invasive Breast Carcinoma With Apocrine Differentiation.
Literature	GENERIF_SUMMARY	TRPS1 outperforms GATA3 in pleural effusions with metastatic breast carcinoma versus mesothelioma.
Literature	GENERIF_SUMMARY	GATA3 mediates doxorubicin resistance by inhibiting CYB5R2-catalyzed iron reduction in breast cancer cells.
Literature	GENERIF_SUMMARY	Loss of function of GATA3 regulates FRA1 and c-FOS to activate EMT and promote mammary tumorigenesis and metastasis.
Literature	GENERIF_SUMMARY	ARID5B, IKZF1, GATA3, CEBPE, and CDKN2A germline polymorphisms and predisposition to childhood acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Gene of the month
Literature	GENERIF_SUMMARY	GATA3 Expression in HPV-associated and HPV-independent Vulvar Squamous Cell Carcinomas
Literature	GENERIF_SUMMARY	Aberrant expression of GATA3 in metastatic adenocarcinoma of the prostate
Literature	GENERIF_SUMMARY	Analysis of GATA3 and FOXA2 expression suggests that downregulation of genes involved in the maintenance of a mature yolk sac tumor phenotype may underlie sarcomatoid transformation.
Literature	GENERIF_SUMMARY	Deregulation of miR-1245b-5p and miR-92a-3p and their potential target gene, GATA3, in epithelial-mesenchymal transition pathway in breast cancer.
Literature	GENERIF_SUMMARY	HepPar1 and GATA-3 Expression in Neuroendocrine Neoplasms
Literature	GENERIF_SUMMARY	The androgen receptor interacts with GATA3 to transcriptionally regulate a luminal epithelial cell phenotype in breast cancer.
Literature	GENERIF_SUMMARY	Recurrent GATA3 P409Afs*99 Frameshift Extension

Main Category	Category	Terms
		Mutations in Sweat-gland Carcinoma With Neuroendocrine Differentiation.
Literature	GENERIF_SUMMARY	Expression of BCL2, TP53, FOXA1, and GATA3 in pTa bladder cancer recurrence.
Literature	GENERIF_SUMMARY	HER2 overexpression in urothelial carcinoma with GATA3 and PPARG copy number gains.
Literature	GENERIF_SUMMARY	Increased level of GATA3-AS1 long non-coding RNA is correlated with the upregulation of GATA3 and IL-4 genes in multiple sclerosis patients.
Literature	GENERIF_SUMMARY	Association of GATA3 expression in triple-positive breast cancer with overall survival and immune cell infiltration.
Pathways	KEGG_PATHWAY	Th1 and Th2 cell differentiation
Pathways	KEGG_PATHWAY	Th17 cell differentiation
Pathways	KEGG_PATHWAY	Parathyroid hormone synthesis, secretion and action
Pathways	KEGG_PATHWAY	Inflammatory bowel disease
Pathways	REACTOME_PATHWAY	Hemostasis
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Cytokine Signaling in Immune system
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Immune System
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	Signaling by Interleukins
Pathways	REACTOME_PATHWAY	Deubiquitination
Pathways	REACTOME_PATHWAY	Ub-specific processing proteases
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	Interleukin-4 and Interleukin-13 signaling
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Transcriptional regulation by RUNX1
Pathways	REACTOME_PATHWAY	ESR-mediated signaling
Pathways	REACTOME_PATHWAY	RUNX1 regulates transcription of genes involved in differentiation of HSCs
Pathways	REACTOME_PATHWAY	Signaling by Nuclear Receptors
Pathways	REACTOME_PATHWAY	Estrogen-dependent gene expression

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages of the Integumentary System
Pathways	REACTOME_PATHWAY	Formation of the nephric duct
Pathways	REACTOME_PATHWAY	Kidney development
Pathways	REACTOME_PATHWAY	Factors involved in megakaryocyte development and platelet production
Pathways	REACTOME_PATHWAY	Developmental Lineages of the Mammary Gland
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Stem Cells
Pathways	WIKIPATHWAYS	Adipogenesis
Pathways	WIKIPATHWAYS	Hair follicle development cytodifferentiation stage 3 of 3
Pathways	WIKIPATHWAYS	Mesodermal commitment pathway
Pathways	WIKIPATHWAYS	Pre implantation embryo
Pathways	WIKIPATHWAYS	Development and heterogeneity of the ILC family
Pathways	WIKIPATHWAYS	White fat cell differentiation
Pathways	WIKIPATHWAYS	Transcriptional cascade regulating adipogenesis
Pathways	WIKIPATHWAYS	GDNF RET signaling axis
Pathways	WIKIPATHWAYS	Familial partial lipodystrophy
Pathways	WIKIPATHWAYS	Th17 cell differentiation pathway
Pathways	WIKIPATHWAYS	Ulcerative colitis signaling
Pathways	WIKIPATHWAYS	Inflammatory bowel disease signaling
Pathways	WIKIPATHWAYS	2q13 copy number variation syndrome
Pathways	WIKIPATHWAYS	16p13 11 copy number variation syndrome
Pathways	WIKIPATHWAYS	Sudden infant death syndrome SIDS susceptibility pathways
Protein Domains	INTERPRO	Znf_GATA
Protein Domains	INTERPRO	Znf_NHR/GATA
Protein Domains	INTERPRO	TF_GATA-2/3
Protein Domains	INTERPRO	Transcription_factor_GATA
Protein Domains	PIR_SUPERFAMILY	TF_GATA-1/2/3

Main Category	Category	Terms
Protein Domains	SMART	ZnF_GATA
Protein Domains	UP_KW_DOMAIN	Repeat
Protein Domains	UP_KW_DOMAIN	Zinc-finger